Changing the particle representation for the viscous step

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Abstract

The Hamiltonian step works with smoothed fields, and the thermodynamic force it provides, b_a, is the fluid velocity smoothed twice. Any viscous step built on b_a damps short wavelengths at most as fast as the smoothing lets through, a factor $R(k)^2$. Here the viscous step is done instead in a particle representation. Each particle is given its own energy e_a, which is held fixed during the viscous step. Momentum is transferred between particles so that the total particle entropy $\sum_a S_a(e_a, \boldsymbol{m}_a)$ increases, and the thermodynamic force is the particle's own $\beta_a = -\partial S_a/\partial \boldsymbol{m}_a$. After the step the particles have the same energies, new momenta and new entropies, and the next Hamiltonian step proceeds from there. We relate the particle energies to the smoothed energy H to quadratic order, and test the deterministic version: on a lattice the damping rate changes from $\Gamma_{\text{phys}} R^2 B$ to $\Gamma_{\text{phys}} B$, in 1D and 2D.

I. TWO REPRESENTATIONS

Each particle carries an entropy s_a and a momentum $\boldsymbol{m}_a$, and sits at $\boldsymbol{x}_a$. The positions do not change during the viscous step.

a. The smoothed representation. The Hamiltonian step uses the smoothed densities at the particle,

$$S_a = \sum_b s_b\, W_{ab}, \qquad \boldsymbol{M}_a = \sum_b \boldsymbol{m}_b\, W_{ab}, \tag{1}$$

and the energy

$$H = \sum_a H_a, \qquad H_a = s_a\, \mathcal{E}(S_a, \boldsymbol{M}_a), \tag{2}$$

where $\mathcal{E}(S, \boldsymbol{M})$ is the energy per unit entropy of a fluid with entropy density S and momentum density $\boldsymbol{M}$. Its thermodynamic force is

$$\mathsf{b}_a = \frac{\partial H/\partial \boldsymbol{m}_a}{\partial H/\partial s_a} = \frac{\dot{\boldsymbol{x}}_a}{\partial H/\partial s_a}. \tag{3}$$

It contains two smoothings: one to form $\boldsymbol{M}$, and one in $\partial H/\partial \boldsymbol{m}_a = \dot{\boldsymbol{x}}_a = \sum_b V_b\, \boldsymbol{v}_b W_{ab}$, because varying $\boldsymbol{m}_a$ changes $\boldsymbol{M}$ at every neighbour.

b. The particle representation. Give each particle the volume

$$V_a = \frac{s_a}{S_a}, \tag{4}$$

so that its own entropy density is the smoothed one. Its own momentum density is $\boldsymbol{m}_a/V_a$. The equation of state then assigns it an energy and, inversely, an entropy:

$$e_a = E_a(s_a, \boldsymbol{m}_a) \equiv V_a\, h\Big(\frac{s_a}{V_a}, \frac{\boldsymbol{m}_a}{V_a}\Big), \qquad s_a = S_a(e_a, \boldsymbol{m}_a) \equiv V_a\, \sigma\Big(\frac{e_a}{V_a}, \frac{\boldsymbol{m}_a}{V_a}\Big). \tag{5}$$

Here $h = S\mathcal{E}$ is the energy density as a function of the entropy and momentum densities, and σ the entropy density as a function of the energy and momentum densities. The two relations are inverse to each other at fixed V_a. The volume is a property of the particle, not a third variable: it is fixed by the configuration and held fixed during the viscous step.

c. Derivatives. From the thermodynamic identity $de = T\,ds + \boldsymbol{v}\cdot d\boldsymbol{m}$ at fixed V_a,

$$\frac{\partial S_a}{\partial e_a} = \frac{1}{T_a}, \qquad \frac{\partial S_a}{\partial \boldsymbol{m}_a} = -\frac{\boldsymbol{v}_a}{T_a} \equiv -\beta_a, \tag{6}$$

where $\boldsymbol{v}_a$ and T_a are the velocity and temperature at the particle's own densities. ($T_a = (e+p)/S$ is the code's density-frame temperature, $T_{\rm rel}/\gamma$.) So β_a is the particle's own thermodynamic force. At rest, to linear order, $\beta_a = \boldsymbol{m}_a/(V_a\, wT)$: proportional to the particle's own momentum, with no smoothing.

II. PARTICLE ENERGIES AND THE SMOOTHED ENERGY

The particle energies eq. (5) and the smoothed energies eq. (2) differ only in the momentum argument. Write the momentum per unit entropy of a particle and its smoothed average as

$$\boldsymbol{\mu}_a = \frac{\boldsymbol{m}_a}{s_a}, \qquad \bar{\boldsymbol{\mu}}_a = \frac{\boldsymbol{M}_a}{S_a} = \sum_b \lambda_{ab}\,\boldsymbol{\mu}_b, \qquad \lambda_{ab} = \frac{s_b W_{ab}}{S_a}, \quad \sum_b \lambda_{ab} = 1. \tag{7}$$

Then

$$e_a = s_a\, f_a(\boldsymbol{\mu}_a), \qquad H_a = s_a\, f_a(\bar{\boldsymbol{\mu}}_a), \qquad f_a(\boldsymbol{\mu}) \equiv \mathcal{E}(S_a, S_a\boldsymbol{\mu}). \tag{8}$$

The function f_a is the energy per unit entropy at the density S_a, as a function of the momentum per unit entropy. Its derivatives are the velocity and the inverse susceptibility,

$$\frac{\partial f_a}{\partial \boldsymbol{\mu}} = \boldsymbol{v}, \qquad \frac{\partial^2 f_a}{\partial \boldsymbol{\mu}\,\partial \boldsymbol{\mu}} = C_a, \tag{9}$$

with C_a positive definite ($C_a = \mathbb{1}/T$ at rest).

a. Convexity. Since f_a is convex, Jensen's inequality gives $f_a(\bar{\boldsymbol{\mu}}_a) \le \sum_b \lambda_{ab} f_a(\boldsymbol{\mu}_b)$. Summing over a with weight s_a, and using that $\sum_a s_a W_{ab}/S_a \simeq 1$ and that S_a varies slowly across a star,

$$H \le \sum_a e_a. \tag{10}$$

The particle energies exceed the smoothed energy by the energy of the momentum fluctuations that the smoothing averages away. This is the energy form of the entropy inequality $S(\underline{P}) \ge \underline{S}$ of `sph1.tex`.

b. Quadratic order. Expand the smoothed energy at particle a, not the particle's own energy. For each neighbour b write $\delta\boldsymbol{\mu}_{ab} = \boldsymbol{\mu}_b - \bar{\boldsymbol{\mu}}_a$, the deviation of its momentum per unit entropy from the local average at a. Then

$$\sum_b \lambda_{ab}\, f_a(\boldsymbol{\mu}_b) = f_a(\bar{\boldsymbol{\mu}}_a) + \bar{\boldsymbol{v}}_a \cdot \sum_b \lambda_{ab}\,\delta\boldsymbol{\mu}_{ab} + \frac{1}{2}\sum_b \lambda_{ab}\,\delta\boldsymbol{\mu}_{ab}\cdot C_a \cdot \delta\boldsymbol{\mu}_{ab} + \dots \tag{11}$$

The linear term vanishes identically, because the average is taken with the same weights that define it:

$$\sum_b \lambda_{ab}\,\delta\boldsymbol{\mu}_{ab} = 0. \tag{12}$$

Multiply by s_a and sum over a. On the left,

$$\sum_a s_a \sum_b \lambda_{ab} f_a(\boldsymbol{\mu}_b) = \sum_b s_b \Big(\sum_a V_a W_{ab} \Big) f(\boldsymbol{\mu}_b) \simeq \sum_b e_b, \tag{13}$$

using the partition of unity $\sum_a V_a W_{ab} \simeq 1$ and $f_a \simeq f_b$ across a star. So, to quadratic order,

$$\boxed{\sum_a e_a - H = \frac{1}{2} \sum_a s_a \sum_b \lambda_{ab}\, \delta\boldsymbol{\mu}_{ab} \cdot C_a \cdot \delta\boldsymbol{\mu}_{ab}.} \tag{14}$$

The right side is the local variance of $\boldsymbol{\mu}$ around each particle, weighted by the inverse susceptibility: manifestly non-negative. This is the energy form of the entropy excess κ of `sph1.tex`.

c. At rest, mode by mode. On a lattice at rest $C = \mathbb{1}/T$, and the variance is the mean of the square minus the square of the mean:

$$\sum_a s_a \sum_b \lambda_{ab}\, |\delta\boldsymbol{\mu}_{ab}|^2 = \sum_b s_b |\boldsymbol{\mu}_b|^2 - \sum_a s_a |\bar{\boldsymbol{\mu}}_a|^2. \tag{15}$$

Smoothing multiplies a mode of wavenumber k by R_k, so $s\bar{\boldsymbol{\mu}} = R\,\boldsymbol{m}$, and with $sT = Vw$,

$$\sum_a e_a - H = \sum_k \big(1 - R_k^2\big)\, \frac{|\boldsymbol{m}_k|^2}{2wV} \;\geq 0. \tag{16}$$

The particle energy counts the full kinetic energy $|\boldsymbol{m}_k|^2/(2wV)$ of every mode; H counts only the fraction R_k^2. Resolved modes ($R \to 1$) contribute nothing; unresolved ones contribute their whole kinetic energy.

III. THE VISCOUS STEP

a. Forward map. From the configuration $(\boldsymbol{x}_a, s_a, \boldsymbol{m}_a)$ after the Hamiltonian step, compute S_a, $V_a = s_a/S_a$ and the particle energies $e_a = E_a(s_a, \boldsymbol{m}_a)$. By construction $S_a(e_a, \boldsymbol{m}_a) = s_a$.

b. Momentum transfer at fixed energies. Hold every e_a and V_a fixed, and transfer momentum between particles. No energy moves between particles during the step. This is the density-frame requirement on the viscous step: the energy flux stays the ideal one, $T^{i0} = T^{0i}$ with T^{0i} the momentum density, as in the Hamiltonian step, and only the stress T^{ij} receives a viscous part. Each transfer conserves the total momentum, and the total particle entropy

$$\mathbb{S} = \sum_a S_a(e_a, \boldsymbol{m}_a) \tag{17}$$

changes by

$$\Delta\mathbb{S} = -\sum_a \beta_a \cdot \Delta\boldsymbol{m}_a + \frac{1}{2} \sum_a \Delta\boldsymbol{m}_a \cdot \frac{\partial^2 S_a}{\partial\boldsymbol{m}_a \partial\boldsymbol{m}_a} \cdot \Delta\boldsymbol{m}_a + \ldots \tag{18}$$

The second term is negative, since S_a is concave.

- *Deterministic.* Take the transfers along the viscous force,

$$\Delta \boldsymbol{m}_a = \Delta t\, \boldsymbol{F}_a, \qquad \boldsymbol{F}_a = -\frac{\partial K}{\partial \beta_a}, \tag{19}$$

  with $K[\beta]$ the bond dissipative function of `bond_dissipation.tex` evaluated with the particle forces. Because K is quadratic, the first term of $\Delta\mathbb{S}$ is exactly $2K\Delta t \geq 0$.

- *Stochastic.* Propose random momentum transfers between pairs, and accept or reject them with the exact $\Delta\mathbb{S}$, so that $\mathbb{S}$ increases on average. The deterministic step is the target for the mean drift.

c. Back map. After the step each particle has the same energy e_a, a new momentum $\boldsymbol{m}'_a$, and the entropy

$$s'_a = S_a(e_a, \boldsymbol{m}'_a). \tag{20}$$

The positions are unchanged, and the next Hamiltonian step proceeds from $(\boldsymbol{x}_a, s'_a, \boldsymbol{m}'_a)$.

d. Energy. The step conserves $\sum_a e_a$ exactly, and the total momentum. It does not conserve H. By eq. (16), $H = \sum_a e_a$ minus the kinetic energy of the unresolved modes. The viscous step damps those modes and turns their kinetic energy into entropy, which H does see, so H increases. This is intended: the kinetic energy of the unresolved modes, which the smoothed representation did not count, becomes heat, which it does.

e. Implementation. `src/changerep.jl`:

- `pointwise_state`: solves the EOS at S_a and momentum density $\boldsymbol{m}_a/V_a$, and returns β_a of eq. (6);
- `particle_energy`: the forward map $e_a = E_a(s_a, \boldsymbol{m}_a)$;
- `eos_from_energy`: the inverse, the entropy density at given energy and momentum densities. In the density frame the momentum density is $\boldsymbol{M} = w\boldsymbol{u}$, and $\varepsilon = w\gamma^2 - p = e + |\boldsymbol{M}|^2/w$, so the rest-frame energy density solves $e = \varepsilon - |\boldsymbol{M}|^2/w(e)$;
- `dissipative_step_changerep_exact!`: forward map, bond force with the particle β, $\Delta\boldsymbol{m}_a = \Delta t\, \boldsymbol{F}_a$, and the exact back map $s'_a = S_a(e_a, \boldsymbol{m}'_a)$;
- `dissipative_step_changerep!`: the same with the entropy updated to first order, $\Delta s_a = -\beta_a \cdot \Delta\boldsymbol{m}_a$.

The bond force accepts any field in place of `b`, so no change to the core code is needed.

IV. DAMPING ON A FROZEN LATTICE

a. Expectation. Put a small momentum wave $\boldsymbol{m}_a \propto \cos kx_a$ on a uniform lattice at rest. Smoothing $\boldsymbol{m}$ into $\boldsymbol{M}$, and $\boldsymbol{v}$ into $\dot{\boldsymbol{x}}$, each multiply it by the smoothing response

$$R(k) = \sum_r a\, W(r) \cos kr, \tag{21}$$

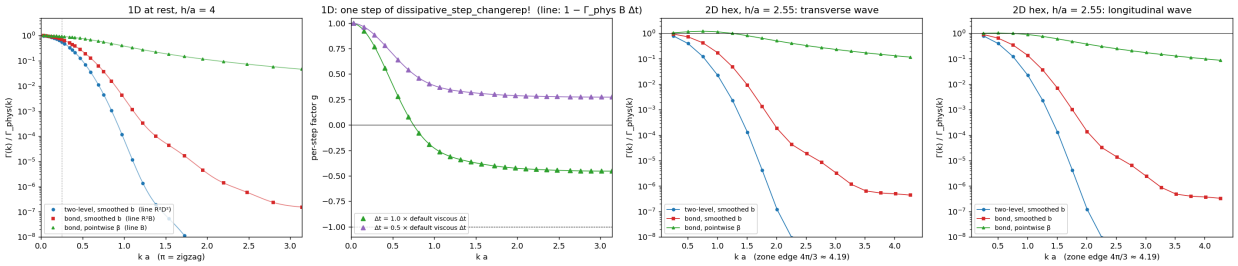


FIG. 1. Damping of a momentum wave on a frozen lattice at rest (`examples/changerep_damping.jl`). First panel: 1D, $h/a = 4$; points are the code, lines the lattice formulas. Second panel: the factor by which one step of `dissipative_step_changerep!` multiplies the mode, at the default viscous Δt and at half of it. Third and fourth panels: 2D hexagonal lattice, $h/a = 2.55$, transverse and longitudinal waves along x.

so $\mathsf{b} \propto R^2 \boldsymbol{m}$, while $\beta \propto \boldsymbol{m}$. The bond force multiplies by $k^2 B(k)$, with the bond response

$$B(k) = \frac{\sum_r \psi(r)\, 2(1 - \cos kr)}{k^2 \sum_r \psi(r)\, r^2}, \tag{22}$$

which tends to one at long wavelengths and is bounded below at short ones. Relative to the physical rate $\Gamma_{\text{phys}} = \kappa k^2 / w$, the damping rates are therefore

$$\frac{\Gamma}{\Gamma_{\text{phys}}} = R^2 B \quad \text{(smoothed } \mathsf{b}\text{)}, \qquad \frac{\Gamma}{\Gamma_{\text{phys}}} = B \quad \text{(pointwise } \beta\text{)}. \tag{23}$$

b. Rates. The measured rate is $\Gamma = -\sum_a \boldsymbol{F}_a \cdot \boldsymbol{m}_a / \sum_a |\boldsymbol{m}_a|^2$ (Fig. 1).

- *1D.* With the pointwise β from the full EOS solve, the rate equals B to 3×10^{-11}, over the whole zone. With the smoothed b it equals $R^2 B$, to the 2×10^{-4} of the discrete kernel normalization. At the zigzag the pointwise rate is $0.046\,\Gamma_{\text{phys}}$, against 1.5×10^{-7} with b and zero for the two-level form.

- *2D.* The same holds for the mean-field bond form on the hexagonal lattice:

ka	two-level, b	bond, b	bond, β
0.25 T	0.79	0.94	1.05
L	0.79	0.90	1.01
1.51 T	1.3×10^{-4}	9.3×10^{-3}	0.81
L	1.3×10^{-4}	7.0×10^{-3}	0.61
4.27 T	2×10^{-14}	4.4×10^{-7}	0.12
L	2×10^{-14}	3.3×10^{-7}	0.087

The pointwise rates tend to one at long wavelengths, overshoot mildly near $ka \approx 0.75$ (to 1.21 for the transverse wave, where the resolved and residual parts of the bond form are both sizable), and fall to about 0.1 at the zone edge. So the change of representation removes the whole R^2 factor, eight orders of magnitude at the zone edge.

c. The time step. One explicit step multiplies a mode by

$$g = 1 - \Gamma\,\Delta t. \tag{24}$$

The measured g agrees with this to 10^{-10}, and the entropy produced equals $2K\Delta t$ to 10^{-15}. But the larger rates matter for Δt. The default viscous step of `compute_dt` is $\Delta t = 0.2\,h^2 w/\kappa$, chosen when the short wavelengths were barely damped. With the pointwise β the largest rate is $\Gamma_{\rm phys}B$ at the zone edge, and

$$\max_k \Gamma\,\Delta t = 1.45 \qquad (1\mathrm{D},\ h/a = 4). \tag{25}$$

The step is still stable ($|g| < 1$), but every mode with $\Gamma\Delta t > 1$ changes sign each step; at the zigzag $g = -0.45$. Halving Δt makes all factors positive, with $g \geq 0.27$. The bound is set by the bond weight: the rate at short wavelengths is about $(\kappa/w)\,2\mu_0/\mu_2$, with $\mu_n = \sum_r \psi(r)|r|^n$, independent of k. In 2D the full viscous Δt exceeds $\Gamma\Delta t = 2$ for the shortest modes, and the explicit step is unstable.

d. The time-step criterion. The particle representation must therefore take smaller time steps: its stronger damping of short wavelengths sets Δt. A bound on the fastest rate follows from Gershgorin's theorem for the bond coefficients,

$$\Gamma_{\max} \leq 4\,\frac{\kappa_b}{w}\,\max_a \sum_b w_{ab}, \qquad \sum_b w_{ab} \leq d_s \frac{\int \psi}{\int \psi\,\rho^2} = \frac{c_d}{h_\psi^2}, \tag{26}$$

with $c_1 = 15/4$ and $c_2 = 7/2$ for $\psi \propto (1 - \rho/2h_\psi)^3$ (the continuum integral includes $\rho \to 0$, which the bond sum omits, so it bounds the lattice sum), and κ_b the largest viscosity the bonds carry: $\kappa(v = 0)$ in 1D and $2\eta + \zeta_{\rm eff}$ in 2D. In 1D at $h/a = 4$ the bound is $15\,\kappa/(wh^2)$, twice the measured rate $7.3\,\kappa/(wh^2)$ of the zigzag mode. For `params.dissipation = :changerep`, `compute_dt` requires $\Gamma_{\max}\Delta t/2 \leq 1$ for each viscous half step of `generic_step!`, so the actual factor is at most about $1/2$ and every mode decays monotonically. This shortens the viscous time step to 0.67 of the previous value in 1D and 0.41 in 2D ($h/a = 2.5$). `test/testchangerep.jl` checks it: all per-step factors of the 1D lattice lie in $[0.52, 1]$, and for a 2D random field the kinetic energy decreases and the entropy increases at every half step.

V. ENERGY BOOKKEEPING OF THE EXACT STEP

`examples/changerep_energy.jl` puts a momentum wave of wavenumber k on the 1D lattice of Sect. IV, with velocity amplitude 10^{-2}, and applies the exact step at half the default viscous Δt until the wave has decayed by 10^{-4}. It records the kinetic energy $\mathrm{KE} = \sum_a |\boldsymbol{m}_a|^2/(2wV_a)$ of the initial wave, the smoothed energy H before and after, the total particle energy, and the entropy produced:

ka	R_k^2	$(1-R_k^2)\,\mathrm{KE}$	$\sum_a e_a - H_0$	$H_{\rm end} - H_0$	drift of $\sum_a e_a$	$T\,\Delta\mathbb{S}/\mathrm{KE}$
0.157	0.893	6.105×10^{-6}	6.118×10^{-6}	6.118×10^{-6}	10^{-11}	1.0000
0.628	0.157	4.829×10^{-5}	4.829×10^{-5}	4.829×10^{-5}	2×10^{-12}	1.0000
1.257	0.001	5.720×10^{-5}	5.720×10^{-5}	5.720×10^{-5}	4×10^{-11}	1.0001
3.142	0	1.145×10^{-4}	1.145×10^{-4}	1.145×10^{-4}	8×10^{-13}	1.0001

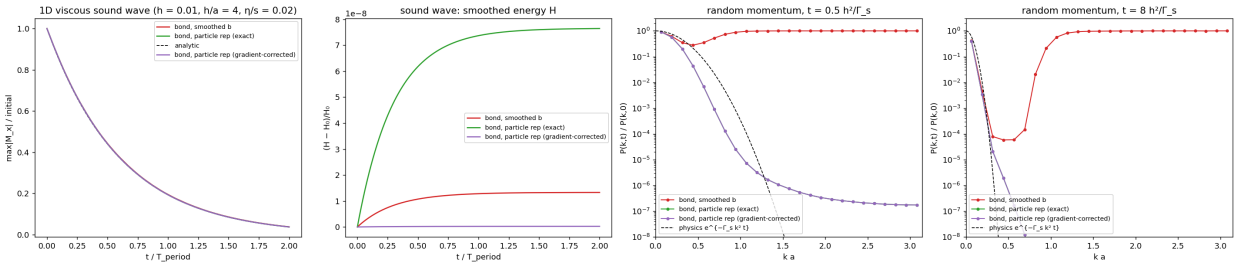


FIG. 2. Full deterministic hydro in 1D (`examples/changerep_hydro.jl`). First two panels: a viscous sound wave ($h = 0.01$, $\eta/s = 0.02$, two periods), amplitude and smoothed energy. Last two panels: the spectrum of a random momentum field ($h = 0.02$, $\eta/s = 0.1$) relative to its initial value, at $t = 0.5$ and 8 in units of h^2/Γ_s, with $\Gamma_s = \kappa/w$.

(At the zigzag $\cos^2 kx_a = 1$ on every particle, so its KE is twice that of the other waves.)

- The excess $\sum_a e_a - H$ agrees with eq. (16) to 2×10^{-3}, the size of the quartic terms at this amplitude.
- $\sum_a e_a$ is conserved to round-off, as it must be: every e_a is held fixed.
- When the wave has decayed, H has grown by exactly the initial excess, $H_{\text{end}} = \sum_a e_a$. The part $R_k^2\,\text{KE}$ of the kinetic energy that H already counted has turned into heat inside H; the part $(1 - R_k^2)\,\text{KE}$ that it did not count has been added to it.
- All of the kinetic energy becomes heat, $T\,\Delta\mathbb{S} = \text{KE}$.

a. Entropy per step. The exact entropy change of a step is smaller than the first-order $2K\Delta t$ by the concave term of Sect. III. For a mode damped at rate Γ, $\Delta\boldsymbol{m} = -\Gamma\Delta t\,\boldsymbol{m}$, and

$$\Delta\mathbb{S} = 2K\Delta t\left(1 - \frac{\Gamma\Delta t}{2}\right), \tag{27}$$

which stays positive for every stable step, $\Gamma\Delta t < 2$. The measured deficit in the first step is 2% at $ka = 0.16$ and 36% at the zigzag, where $\Gamma\Delta t = 0.73$, as this formula predicts. Over the whole decay the deficits do not matter: the energy is exact, so the heat produced is exactly the kinetic energy removed.

VI. FULL DETERMINISTIC RUNS

`examples/changerep_hydro.jl` alternates viscous and Hamiltonian steps, $D(\Delta t/2)\,H(\Delta t)\,D(\Delta t/2)$, on the 1D lattice ($h/a = 4$), with the viscous step D done with the bond form on the smoothed `b`, as in `generic_step!`, or in the particle representation with the exact or the gradient-corrected map of Appendix A (Fig. 2). All runs here use the corrected equation of state of Appendix B. Each half step uses $\Delta t/2$, so at the default viscous Δt every mode has $\Gamma\Delta t/2 \le 0.73$ and decays monotonically.

a. Viscous sound wave. The wave is resolved, $kh = 0.13$. After two periods:

	smoothed b	particle, exact	particle, gradient-corrected
amplitude (analytic 0.03713)	0.03775	0.03719	0.03719
$\Delta S/S$	1.27×10^{-5}	1.27×10^{-5}	1.27×10^{-5}
$\Delta H/H$	1.3×10^{-8}	7.6×10^{-8}	2.5×10^{-10}

Both particle versions follow the analytic decay more closely than the smoothed b. The exact map adds energy to H, the gradient part of the excess released as the wave decays; the corrected map does not.

b. Random momentum field. Each particle starts with a random momentum (velocity $\sim 10^{-3}$) on a uniform fluid, so most of the kinetic energy is in unresolved modes.

- With the smoothed b, the unresolved modes are not damped: for $ka \gtrsim 1$ the spectrum stays at its initial value, and after $8h^2/\Gamma_s$ 72% of the kinetic energy is left.
- In the particle representation every mode is damped. After $8h^2/\Gamma_s$ only 1% of the kinetic energy is left, and the resolved modes follow the physical decay.
- At early times the intermediate modes, $0.5 \lesssim ka \lesssim 1.3$, decay faster than the physical sound damping $e^{-\Gamma_s k^2 t}$. Their pressure force passes through the smoothing, so they barely oscillate. A propagating sound wave keeps half its energy as kinetic energy, which alone is damped; a mode that does not oscillate keeps all of it, so its momentum spectrum decays at twice the rate.

The energy bookkeeping is as in Sect. V:

viscous step	$T\,\Delta S$	ΔH	kinetic energy removed
smoothed b	1.14×10^{-7}	3.4×10^{-9}	3.2×10^{-7}
particle, exact	1.12×10^{-6}	1.01×10^{-6}	1.12×10^{-6}
particle, gradient-corrected	1.06×10^{-6}	0.95×10^{-6}	1.12×10^{-6}

With the smoothed b the scheme conserves H: the heat produced is kinetic energy that H already counted. (The kinetic energy removed exceeds $T\Delta S$ there, because in the resolved modes, which oscillate, the Hamiltonian step also turns kinetic energy into the energy of entropy fluctuations.) In the particle representation the heat produced is the kinetic energy removed, and H grows by the part of it that the smoothing did not count. The gradient correction leaves this almost unchanged: the energy of the random field is in the residual, not in a smooth gradient, and it is still turned into heat.

VII. TWO DIMENSIONS

The same comparison in 2D, with the standard tests run unchanged and only the viscous step replaced: the bond form on the smoothed b, as in `generic_step!`, or the step in the particle representation, with the exact map or with the gradient-corrected map of Appendix A. All runs use the corrected equation of state of Appendix B.

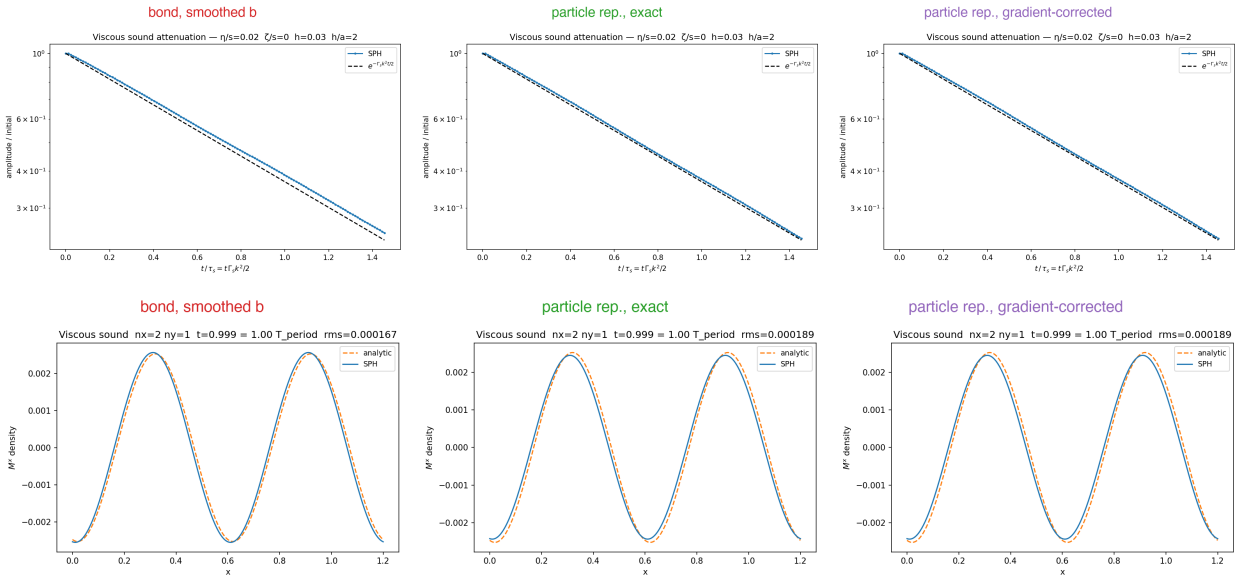


FIG. 3. 2D viscous sound wave (`testsound_viscous.jl`), with the viscous step on the smoothed `b` (left), in the particle representation with the exact map (centre), and with the gradient-corrected map (right). Top: amplitude against $e^{-\Gamma_s k^2 t/2}$. Bottom: M^x along x after one period.

a. Viscous sound wave. `examples/changerep_sound.jl` runs `testsound_viscous.jl` ($h = 0.03$, $h/a = 2$, $\eta/s = 0.02$, one period; Fig. 3). With the smoothed `b` the wave is underdamped throughout; in the particle representation the amplitude follows the analytic decay closely:

	smoothed `b`	particle, exact	particle, gradient-corrected
amplitude (analytic 0.2337)	0.2470	0.2363	0.2363
$\Delta S/S$	1.172×10^{-5}	1.209×10^{-5}	1.172×10^{-5}
$\Delta H/H$	9.8×10^{-8}	5.1×10^{-7}	1.9×10^{-8}

The gradient correction keeps the accuracy of the particle representation and removes its energy gain; the entropy produced is then that of the smoothed scheme. On the final profile all runs share a small phase lead, which comes from the smoothed sound speed rather than the viscous step.

b. Gaussian. `examples/changerep_gaussian_lines.jl` evolves the Gaussian of `testgaussian.jl` ($h = 0.015$, $\eta/s = 0.002$) with the smoothed `b` and with the gradient-corrected particle representation, and records the energy density along the two axes through the centre at several times (Fig. 4). The two schemes give the same flow at every time. The differences are at most 0.004, against a peak of 1.1, and have no systematic sign; at $t = 2.2$ they are below 0.0006. (With the exact map, the late-time energy density was higher by about 0.002 almost everywhere: the energy gain discussed next, spread over the fluid as heat.)

c. Long-time energy. The leapfrog conserves a nearby (shadow) Hamiltonian, not H itself, so H oscillates; what matters is whether it drifts. For resolved viscous hydro every scheme should give, as $h \to 0$, a constant $H(t)$ and the same growing $S(t)$.

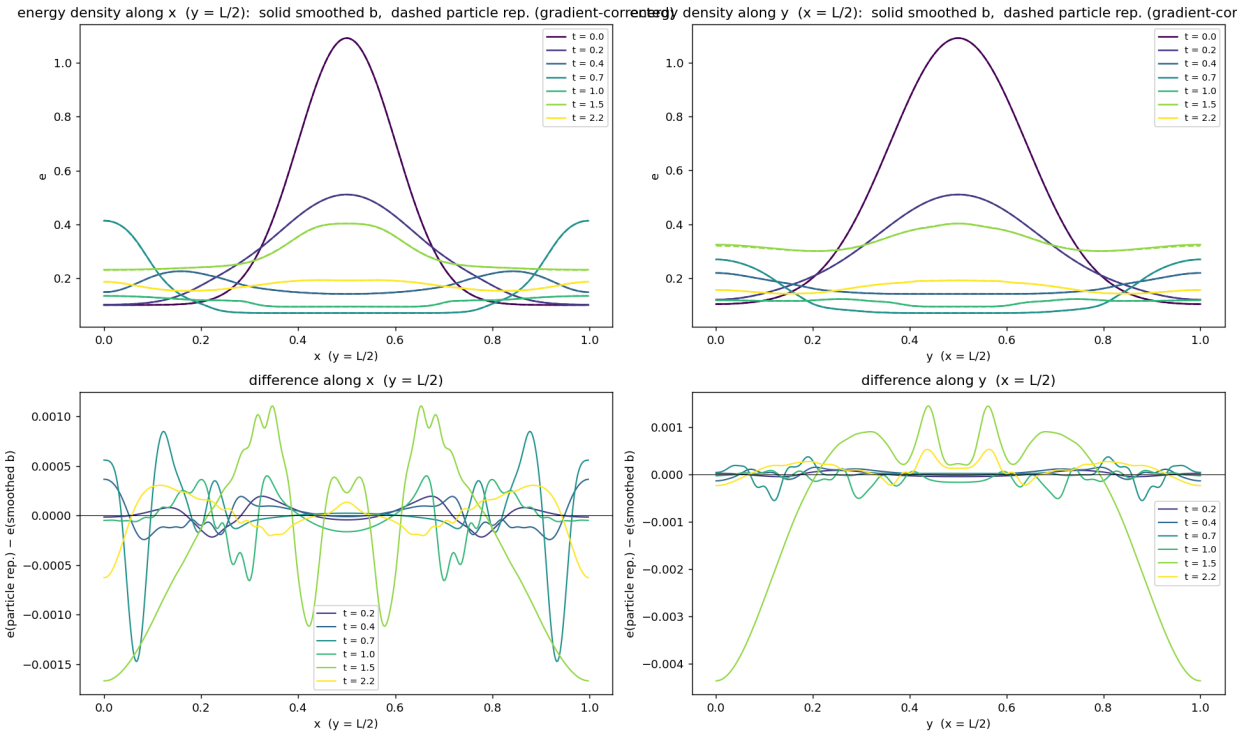


FIG. 4. 2D Gaussian: energy density along x (left) and y (right) through the centre, at $t = 0$ to 2.2. Top: smoothed b (solid) and particle representation with the gradient-corrected map (dashed). Bottom: their difference.

`examples/changerep_energy_drift.jl` records $H(t)$ and $S(t)$ after every full step for the Gaussian at $h = 0.03$ up to $t = 4$ (Fig. 5):

- The two-level form and the bond form on the smoothed b give the same $H(t)$, constant to 3×10^{-4}.
- With the exact map, H grows steadily, to 2.7×10^{-3} at $t = 4$, and the entropy grows faster.
- With the gradient-corrected map, $H(t)$ and $S(t)$ follow the smoothed schemes.

d. Convergence. `examples/changerep_convergence.jl` repeats this at $h = 0.045$, 0.03, 0.0225 and 0.015 up to $t = 3$ (Fig. 6, and Fig. 7 in the appendix).

- *The particle representation works without any correction, provided h is small enough.* With the exact map the late-time energy gain is 4.9, 2.3, 1.3 and 0.58×10^{-3}: it converges as h^2. The entropy produced converges to that of the smoothed schemes, 4.27% against 4.20–4.22% at $h = 0.015$.
- The gain is secular: the excess energy is proportional to the entropy produced, $\Delta H_{\text{excess}}/H \simeq c\,h^2\,\Delta S/S$ with $c/h^2 \approx 60$ (third panel of Fig. 6). Every unit of viscous heat comes with an extra energy of relative size $\propto h^2$: the smooth-gradient part of the excess, released along with the physical dissipation.

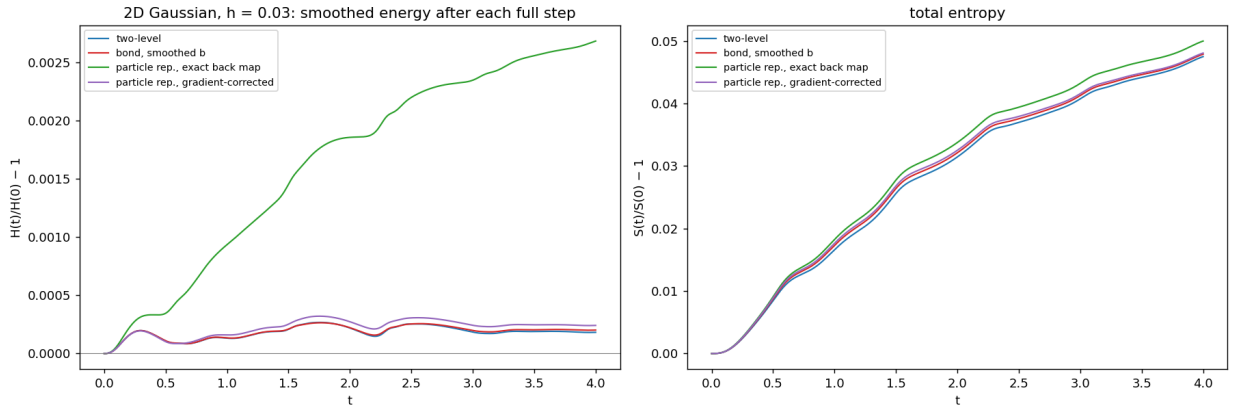


FIG. 5. 2D Gaussian at $h = 0.03$: smoothed energy (left) and total entropy (right) after each full step.

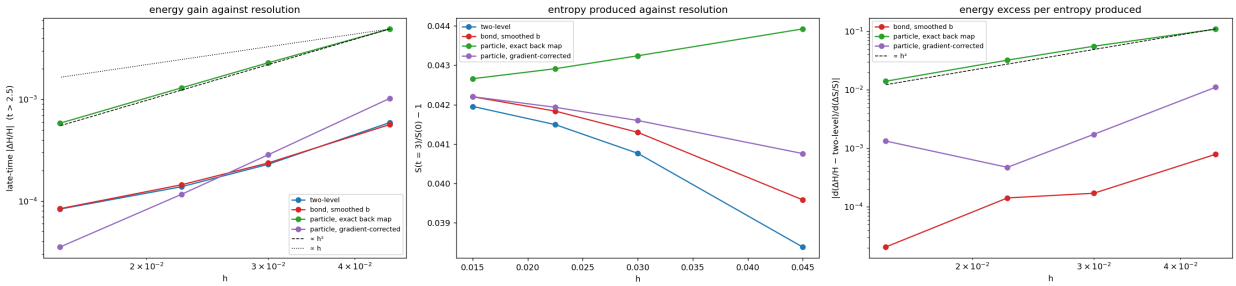


FIG. 6. 2D Gaussian, convergence in h. Left: late-time energy gain ($t > 2.5$). Centre: entropy produced at $t = 3$. Right: slope of the energy excess over the two-level form against the entropy produced.

- The gradient-corrected map removes the lion's share of this secular gain: the excess per entropy produced drops by a factor 10 at $h = 0.045$, 30 at $h = 0.03$, and to the level of the smoothed schemes at smaller h.

VIII. OPEN POINTS

The gradient-corrected particle representation is now the default viscous step of the code, `params.dissipation = :changerep`; `test/testchangerep.jl` checks the conversion map, the rates of Sect. IV, the gradient correction and conservation.

- *Speed.* The gradient-corrected step is several times slower than the bond step on the smoothed `b`: each half step rebuilds every particle's kernel star and fits G_a twice, with a pseudo-inverse. Closed-form 2×2 fits, reused neighbour lists and a single pass over the stars should bring it close to the cost of the bond step.

- *The force of the corrected map.* The gradient correction adds a higher-derivative term to the thermodynamic force, eq. (A8); it is not yet included in the force.

- *Time step.* An implicit or exponential update of the bond step would remove the time-step restriction of Sect. IV, which the code now enforces (below).
- *Stochastic step.* Random momentum transfers between pairs, accepted with the exact $\Delta\mathbb{S}$; their mean drift should reproduce the deterministic rates of Sect. IV.

Appendix A: A gradient-corrected conversion map

a. The smooth part of the excess. By eq. (14), the particle energies exceed H by the local variance of $\boldsymbol{\mu}$ around each particle. For a smooth flow the deviations are not fluctuations but the gradient across the star. With $\bar{\boldsymbol{r}}_a = \sum_b \lambda_{ab}\boldsymbol{r}_{ab}$ the centre of the star (zero on a symmetric one),

$$\delta\boldsymbol{\mu}_{ab} \simeq \nabla\boldsymbol{\mu}\cdot\big(\boldsymbol{r}_{ab} - \bar{\boldsymbol{r}}_a\big), \tag{A1}$$

and the excess contains

$$g_a = \frac{1}{2}\, s_a\, (G_a)_{ij}\, C_a^{ik}\, (G_a)_{kl}\, Q_a^{jl}, \qquad Q_a^{jl} = \sum_b \lambda_{ab}\,(r_{ab} - \bar{r}_a)^j (r_{ab} - \bar{r}_a)^l, \tag{A2}$$

with G_a the gradient of $\boldsymbol{\mu}$ at a. Q_a is the second moment of the kernel, of order h^2, so even a perfectly resolved flow has an excess of order $h^2|\nabla\boldsymbol{\mu}|^2$ per unit entropy. The uncorrected step holds every e_a fixed, so as viscosity reduces the gradient it adds $\sum_a \Delta g_a$ to H. That is energy the smoothed representation already counted, as part of the resolved flow; ordinary viscosity would leave H unchanged.

b. The corrected map. Treat the linear part of the variation as resolved, and only the residual as the particle's own fluctuation. Take G_a to be the least-squares gradient of $\boldsymbol{\mu}$ on the star of a with the kernel weights λ_{ab} (the construction of `bond_dissipation.tex`, with these weights in place of the bond weights), so that the residual

$$\delta\boldsymbol{\mu}'_{ab} = \boldsymbol{\mu}_b - \bar{\boldsymbol{\mu}}_a - G_a\big(\boldsymbol{r}_{ab} - \bar{\boldsymbol{r}}_a\big) \tag{A3}$$

is orthogonal to the linear part, $\sum_b \lambda_{ab}\,\delta\boldsymbol{\mu}'_{ab}(\boldsymbol{r}_{ab} - \bar{\boldsymbol{r}}_a)^T = 0$. The cross term in the variance then vanishes for any C_a, and the excess splits exactly, at quadratic order, into $\sum_a g_a$ and the variance of the residual. The step becomes:

1. *Forward:* $e'_a = e_a - g_a$, with e_a the energy the equation of state assigns to $(s_a, \boldsymbol{m}_a)$ at volume V_a.
2. *Transfer:* move momentum between particles at fixed e'_a.
3. *Back:* recompute the fit G'_a from the new momenta, hence g'_a, and set

$$s'_a = S_a\big(e'_a + g'_a,\ \boldsymbol{m}'_a\big). \tag{A4}$$

The gradient energy moves with the resolved flow, as H treats it, and only the variance of the residual is turned into heat. For a resolved flow the step then conserves H, like ordinary viscosity, while the particle-scale fluctuations are still released. It is the resolved-plus-residual split of the bond form, applied to the conversion map.

c. Maximizing entropy at fixed constraints. Both versions of the step increase the total particle entropy at fixed constraints.

- *Uncorrected.* Every e_a is held fixed, and the step increases

$$\mathbb{S} = \sum_a S_a(e_a, \boldsymbol{m}_a) \tag{A5}$$

  over the momenta, at fixed total momentum. The constraints are local, and the force conjugate to $\boldsymbol{m}_a$ is $\beta_a = -\partial S_a/\partial \boldsymbol{m}_a$.

- *Corrected.* The constraint becomes

$$e'_a = e_a - g_a[\boldsymbol{m}] = \text{const}, \tag{A6}$$

  and g_a depends on the neighbours' momenta through the fitted gradient G_a. The constraint now contains a square-gradient term, with a coefficient set by the kernel's second moment, of order h^2. Equivalently the step increases

$$\mathbb{S}' = \sum_a S_a\big(e'_a + g_a[\boldsymbol{m}],\ \boldsymbol{m}_a\big) \qquad \text{at fixed } e'_a. \tag{A7}$$

Differentiating with respect to $\boldsymbol{m}_c$, the derivative term adds to the thermodynamic force

$$\beta_c^{\text{eff}} = \beta_c - \sum_a \frac{1}{T_a}\frac{\partial g_a}{\partial \boldsymbol{m}_c}. \tag{A8}$$

Since g_a is quadratic in the gradient, the correction is a discrete divergence of $C\,\nabla\boldsymbol{\mu}$: a higher-derivative correction of order h^2. Summed over particles, the corrected constraint is

$$\sum_a e'_a = \sum_a e_a - \sum_a g_a \simeq H + (\text{variance of the residual}), \tag{A9}$$

so the derivative term moves the constraint from $\sum_a e_a$ towards the resolved energy H plus the energy of the particle-scale fluctuations.

d. Implementation. `dissipative_step_changerep_grad!` in `src/changerep.jl`:

1. smooth, and for each particle build its star with the kernel weights λ_{ab}, the centred second moment Q_a, the fitted gradient G_a of $\boldsymbol{\mu}$, and C_a of eq. (A11) at the smoothed state;
2. set $e'_a = e_a - g_a$;
3. compute the particle forces β_a, the bond force, and $\Delta\boldsymbol{m}_a = \Delta t\,\boldsymbol{F}_a$;
4. refit G'_a from the new momenta on the same stars, and set $s'_a = S_a(e'_a + g'_a, \boldsymbol{m}'_a)$.

The force uses the uncorrected β_a; the derivative term of β^{eff} is not included.

e. The inverse susceptibility. $C_a = \partial^2 f_a/\partial\boldsymbol{\mu}\,\partial\boldsymbol{\mu} = S\,\partial\boldsymbol{v}/\partial\boldsymbol{M}|_S$ has a closed form. At fixed $S = s\gamma$ the entropy per volume changes as $ds/s = -d\gamma/\gamma$, so with $ds/de = s/w$ the rest-frame energy density changes as $de = -w\,d\ln\gamma$, and $dw = (1 + c_s^2)\,de$. Differentiating $\boldsymbol{M} = w\gamma^2\boldsymbol{v}$, with $d\gamma = \gamma^3\,\boldsymbol{v}\cdot d\boldsymbol{v}$,

$$\frac{\partial \boldsymbol{M}}{\partial \boldsymbol{v}}\Big|_S = w\gamma^2\big[\mathbb{1} + \gamma^2(1 - c_s^2)\,\boldsymbol{v}\boldsymbol{v}^T\big], \tag{A10}$$

and inverting,

$$C = S\,\frac{\partial \boldsymbol{v}}{\partial \boldsymbol{M}}\Big|_S = \frac{1}{T\gamma^2}\left[\mathbb{1} - \frac{(1 - c_s^2)\,\boldsymbol{v}\boldsymbol{v}^T}{1 - c_s^2 v^2}\right], \qquad T = \frac{e + p}{S} = \frac{T_{\rm rel}}{\gamma}. \tag{A11}$$

Transverse to $\boldsymbol{v}$ it is $1/(T\gamma^2)$, along $\boldsymbol{v}$ it is $1/[T\gamma^4(1 - c_s^2 v^2)]$, and at rest it is $\mathbb{1}/T$. It agrees with a finite-difference evaluation through the equation of state to 10^{-10} up to $v = 0.83$.

f. A first check. For a smooth shear flow on the 2D lattice ($\boldsymbol{\mu} \propto \sin 2\pi x\,\hat{y}$, $h = 0.05$), the gradient energy $\sum_a g_a$ accounts for 98.8% of the excess $\sum_a e_a - H$. In one viscous step, the change of H relative to the heat produced, $\Delta H/(T\Delta S)$, is 0.027 for the uncorrected exact step and 0.0004 for the corrected one.

g. Convergence. Fig. 7 shows $H(t)$ and $S(t)$ for the Gaussian at four resolutions, for the two smoothed schemes and the two maps of the particle representation. The late-time energy gain, $\Delta H/H$ for $t > 2.5$:

h	two-level	bond, smoothed b	particle, exact	particle, gradient-corrected
0.045	5.9×10^{-4}	5.7×10^{-4}	4.9×10^{-3}	1.0×10^{-3}
0.03	2.3×10^{-4}	2.4×10^{-4}	2.3×10^{-3}	2.9×10^{-4}
0.0225	1.4×10^{-4}	1.4×10^{-4}	1.3×10^{-3}	1.2×10^{-4}
0.015	8.3×10^{-5}	8.4×10^{-5}	5.8×10^{-4}	3.5×10^{-5}

With the gradient correction, the energy gain is that of the smoothed schemes, or smaller at fine resolution, and the entropy produced agrees with the bond form on the smoothed b: 4.220% against 4.220% at $h = 0.015$. The slope of the excess energy against the entropy produced (Fig. 6) is 1.1×10^{-2}, 1.7×10^{-3}, -0.5×10^{-3} and -1.3×10^{-3}, against 0.109, 0.056, 0.032 and 0.014 for the exact map. The correction is not exact—the force still uses the uncorrected β_a, and g_a captures the smooth excess only to quadratic order—but it removes most of the secular gain.

Appendix B: The equation-of-state inversion

The energy studies of this note exposed an error in `solve_eos!`, which was corrected before the runs reported here (`test/testeos_consistency.jl`).

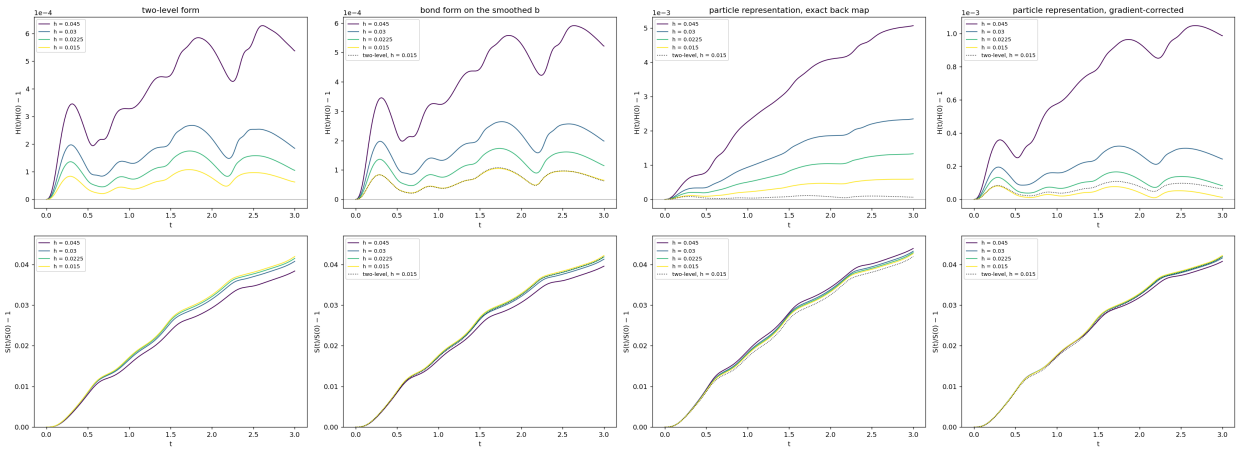


FIG. 7. 2D Gaussian, $H(t)$ (top) and $S(t)$ (bottom) at $h = 0.045$ to 0.015 for the two-level form, the bond form on the smoothed `b`, and the particle representation with the exact and the gradient-corrected map. Each panel has its own vertical scale. Dotted: two-level at $h = 0.015$.

a. The error. The Hamiltonian uses the energy density $h = (e+p)\gamma^2 - p$ as a function of the entropy density $S = s\gamma$ and the momentum density. This energy is conjugate to the density-frame momentum

$$M^i = (e+p)\gamma^2 v^i = (e+p)\gamma\, u^i, \tag{B1}$$

for which $\partial h/\partial \boldsymbol{M} = \boldsymbol{v}$ and $\partial h/\partial S = (e+p)/S$, and for which the post-convergence expression for $\mathcal{F}/S$ follows from the Euler relation $h + p = TS + \boldsymbol{v}\cdot\boldsymbol{M}$. The solver instead took $\boldsymbol{u} = \boldsymbol{M}/(e+p)$, dropping a factor γ (the same step was missing in `sph2.tex`, §II.4). For a moving fluid the energy was then not conjugate to the momentum: the Hamiltonian velocity $\dot{\boldsymbol{x}}_a$ and $T_{\rm eff}$ were not the derivatives of H, wrong at $O(v^2)$ (by 10–20% at $v \approx 0.2$–0.35).

b. Consequences. Neither the leapfrog nor any viscous step could conserve H in a moving flow; even the two-level step changed H at first order in Δt. The resolved, slow flows of the standard tests were hardly affected, since the error is $O(v^2)$. The energy studies of this note were the first to require exact bookkeeping in a strongly moving flow. With the corrected solver the oscillation of H in the Gaussian drops from 8×10^{-3} to 3×10^{-4}, the exact and first-order entropy updates of the particle representation agree, and the energy gain converges as h^2.